

JEE MAIN REVISION MASTER: CHEMICAL KINETICS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 34 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Rate Law, Order & Integrated Rate Equations	25 PYQs	Rate Laws
High	Arrhenius Equation & Activation Energy	9 PYQs	Temperature & Catalysis

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Rate Law, Order & Integrated Rate Equations	$\text{Zero: } [A] = [A]_0 - kt, \quad t_{1/2} = \frac{[A]_0}{2k}, \quad \text{First: } k = \frac{2.303}{t} \log \frac{[A]_0}{[A]}, \quad t_{1/2} = \frac{0.693}{k}$
Arrhenius Equation & Activation Energy	$k = A e^{-E_a/RT}, \quad \ln k = \ln A - \frac{E_a}{RT}, \quad \log_{10} \left(\frac{k_2}{k_1} \right) = \frac{E_a}{2.303R} \left(\frac{T_2 - T_1}{T_2 T_1} \right)$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Rate Law, Order & Integrated Rate Equations (Very High Yield — 25 PYQs)

Core Principle: Differential and integrated rate laws, half-life formulas, and overall reaction order determined from initial rates.

■ **Caution / JEE Pitfall:** For a first-order reaction, half-life $t_{1/2} = (0.693 / k)$ is strictly INDEPENDENT of initial concentration $[A]_0$.

■ **Speed Trick:** For an n^{th} order reaction: $t_{1/2} \propto \frac{1}{[A]_0^{n-1}}$. Units of rate constant $k = \text{mol}^{1-n} \text{L}^n \text{s}^{-1}$.

Representative Source PYQs:

- [2026 Shift] Q#2 (MCQ): "For a 1st order change $R \rightarrow P$, the concentration of Reactant R changes from 0.1 M to 0.025 M in 40 minutes. The rate of re..." [Ans: N/A]
- [2026 Shift] Q#4 (MCQ): "Which one of the following graph is not applicable for a 1st order reaction ($R \rightarrow P$)?..." [Ans: N/A]
- [2025 Shift] Q#2 (MCQ): "Half-life of a first order reaction is 20 seconds and initial concentration of reactant is 0.2 M. The concentration of reactant left after ..." [Ans: N/A]

3.2 Arrhenius Equation & Activation Energy (High Yield — 9 PYQs)

Core Principle: Temperature dependence of reaction rates, activation energy E_a , Arrhenius plot ($\ln k$ vs $1/T$), and catalyst energy profiles.

■ **Caution / JEE Pitfall:** A positive catalyst increases rate by LOWERING activation energy E_a ; it does NOT alter the equilibrium constant K_{eq} or ΔH !

■ **Speed Trick:** Plot of $\log_{10} k$ vs $(1/T)$ gives a straight line with slope = $-(E_a / 2.303 R)$ and y-intercept = $\log_{10} A$.

Representative Source PYQs:

- **[2026 Shift] Q#1 (MCQ):** "The activation energy for the reaction $X \rightarrow Y$ is 150 kJ mol^{-1} . The change in enthalpy for the above reaction is -135 kJ mol^{-1} ." [Ans: N/A]
- **[2026 Shift] Q#3 (MCQ):** "For a reaction having three steps, the overall rate constant is $k = \frac{k_1 k_2 k_3}{k_{-1} k_{-2}}$. The values of E_{a1}, E_{a2} and E_{a3} are..." [Ans: N/A]
- **[2025 Shift] Q#1 (MCQ):** "Catalysts are used to increase the rate of a chemical reaction. Because it..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

The activation energy for the reaction $X \rightarrow Y$ is 150 kJ mol^{-1} . The change in enthalpy for the above reaction is -135 kJ mol^{-1} . Then the activation energy for $Y \rightarrow X$ is

- (A) 280 kJ mol^{-1}
- (B) 285 kJ mol^{-1}
- (C) 270 kJ mol^{-1}
- (D) 15 kJ mol^{-1}

Correct Answer: None

Solution Analysis: For the reaction $X \rightarrow Y$ activation energy is $E_a(\text{forward}) = 150 \text{ kJ mol}^{-1}$ and enthalpy change is $\Delta H = -135 \text{ kJ mol}^{-1}$. Since ΔH is negative, the reaction is exothermic. This means product Y has lower energy than reactant X by 135 kJ mol^{-1} . For the reverse reaction, $Y \rightarrow X$ the activation energy will be the energy needed to go from Y up to the activated complex. Using the relation: $E_a(\text{reverse}) = E_a(\text{forward}) - \Delta H$. Substitute the values: $E_a(\text{reverse}) = 150 - (-135) = 285 \text{ kJ mol}^{-1}$.

Benchmark Problem 2: KCET 2026 (Q#2)

For a 1st order change $R \rightarrow P$, the concentration of Reactant R changes from 0.1 M to 0.025 M in 40 minutes. The rate of reaction when the concentration of R is 0.01 M is _____

- (A) $1.73 \times 10^{-5} \text{ M min}^{-1}$
- (B) $3.47 \times 10^{-4} \text{ M min}^{-1}$
- (C) $3.47 \times 10^{-5} \text{ M min}^{-1}$
- (D) $1.73 \times 10^{-4} \text{ M min}^{-1}$

Correct Answer: None

Solution Analysis: For a first order reaction, $k = \frac{2.303}{t} \log \frac{[R]_0}{[R]}$. Here, $[R]_0 = 0.1 \text{ M}$, $[R] = 0.025 \text{ M}$, $t = 40 \text{ min}$. So, $k = \frac{2.303}{40} \log \frac{0.1}{0.025} = \frac{2.303}{40} \log 4 = 0.0347 \text{ min}^{-1}$. For a first order reaction, the rate is $\text{Rate} = k[R]$. When the concentration of R is 0.01 M , $\text{Rate} = 0.0347 \times 0.01 = 3.47 \times 10^{-4} \text{ M min}^{-1}$. So, the correct option is $3.47 \times 10^{-4} \text{ M min}^{-1}$ Option B.